



Indian Institute of Information Technology, Surat

End Semester Examination- Dec.- 2025

B.Tech. (3rd Sem)

AS 305 – Probability and Statistical Analysis

Date : 11/12/2025

Time Allowed: 3:00 hrs.

Reg. No.

224556

Max. Marks: 100

Instructions: Answer all the questions.

1. (a) Let $P(B) > 0$. Prove that the conditional probability $P[A|B]$ satisfies axioms (i)-(iii) of the probability definition. (5)
- (b) Find a 95% confidence interval estimating the mean height of the 1546 male students at a university by taking a sample of size 100. (Assume the mean of the sample, is 67.45 and that the standard deviation of the sample, is 2.93 inches.) ($Z_c = 1.96$ two tailed, $Z_c = 1.645$ one-tailed) (5)
2. (a) If X is a discrete random variable and $E[X^2] = 0$, show that $P(X = 0) = 1$. Deduce that if $V(X) = 0$ then $P(X = \mu) = 1$, where $\mu = E[X]$. (5)
- (b) A symmetric die is thrown 720 times. Use Chebyshev's inequality to find the lower bound for the probability of getting 100 to 140 sixes. (5)
3. (a) Let X be a random variable with moment generating function $0.125(1 + e^t)^3, t \in \mathbb{R}$. Then find $V(2X - 1)$. (5)
- (b) Bus arrives at a specified stop at 15-minute intervals starting at 7:00 AM. That is, they arrive at 7:00, 7:15, 7:30, 7:45, and so on. If a passenger arrives at the stop at a time that is uniformly distributed between 7 and 7:30, find the probability that
 - (i) less than 5 minutes for a bus,
 - (ii) more than 10 minutes for a bus.

Let X and Y be independently distributed random variable, and

$$P[X = i] = P[Y = i] = \frac{1}{n}, \forall i = 1, \dots, n.$$

Find $P[X < Y]$, $P[X = Y]$, and $P[X > Y]$.

(10)

4. (a) State and prove the conditions under which the Poisson distribution serves as an approximation to the Binomial distribution. (10)
- (b) If the joint density function of random variables X and Y is given by

$$f_{X,Y}(x,y) = \begin{cases} xy^2 + \frac{x^2}{8}, & 0 < x < 2, 0 < y < 1; \\ 0, & \text{elsewhere.} \end{cases}$$

Find (i) $P(X > 1)$, (ii) $P\left(Y < \frac{1}{2}\right)$, (iii) $P\left(X > 1 | Y < \frac{1}{2}\right)$, (iv) $P\left(Y < \frac{1}{2} | X > 1\right)$. (10)

- (c) A market research analyst wants to determine whether the market penetration of a particular brand is uniform across different parts of the country. The country is divided into four regions, and a sample of 100 consumers is taken from each region. The survey results are shown below: ($\chi^2_{(3,0.05)} = 7.815$)

	Region				
	NE	NW	SE	SW	Total
Purchased the brand	40	55	45	50	190
Did not purchase	60	45	55	50	210

Using χ^2 test, find out if the brand is unevenly distributed throughout the country.

(10)

5. (a) Let $X_1, X_2, \dots, X_n$ be an i.i.d. sample from the uniform distribution

$$X_i \sim U(\theta_1, \theta_2), \quad \theta_1 < \theta_2.$$

Then derive estimators for θ_1 and θ_2 Using the method of moments (MoM).

(5)

- (b) Let $X_1, \dots, X_n$ be an i.i.d. sample from the Negative Binomial distribution with parameters $r > 0$ and $0 < p < 1$, where the pmf (count of failures before r successes) is

$$P(X = x) = \binom{x+r-1}{x} (1-p)^r p^x, \quad x = 0, 1, 2, \dots,$$

then derive the maximum likelihood estimator (MLE) of p when r is known.

(5)

- (c) Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables with pdf

$$f(x; \theta) = \theta x^{\theta-1}, \quad 0 < x < 1, \theta > 0.$$

Define the estimator $T_n = \frac{1}{n} \sum_{i=1}^n (-\ln X_i)$. Prove or disprove that T_n is unbiased estimator of θ .

(5)

6. (a) In a large city A, 20% of a random sample of 900 school children had defective eye-sight. In other city B, 15% of a random sample of 1600 children had the same defect. Is this difference between the two proportions significant at 5% level of significance? ($Z_c = 1.96$ two tailed, $Z_c = 1.645$ one-tailed)

(5)

- (b) A biologist records the bacterial population (in millions) at different temperatures and wants to model the data using $Y = ae^{bX+cX^2}$. The collected data are:

Temperature X ($^{\circ}C$)	1	2	3	4	5
Population Y (millions)	2.5	4.2	8.5	14.9	22.3

- Determine estimates of the parameters a , b , and c .
- Find the mean square error for the least squares principle.
- Find the correlation between X and Y .

(5+3+2)

Best wishes